



Choroidal melanoma

Hypofractionated stereotactic photon radiotherapy of posteriorly located choroidal melanoma with five fractions at ten Gy – Clinical results after six years of experience



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ABSTRACT

Purpose: To evaluate long-term safety and efficacy of hypofractionated stereotactic photon radiotherapy with 5 fractions at 10 Gy each in patients with centrally located choroidal melanoma.

Materials and Methods: Ninety-one patients with centrally located choroidal melanoma were treated stereotactically at a linear accelerator with 6 MV photon beams with 5 fractions at 10 Gy each. Examinations were performed at baseline and every 3 months in the first 2 years, then every 6 months until 5 years and yearly thereafter. Median follow-up was 37.8 months (IQR 19.2–49.9). They included visual acuity assessment, routine ophthalmological examinations with fundoscopy, echography for measurement of tumor dimensions, medical examinations and, if necessary, fluorescein angiography.

Results: Initial tumor base diameters, height and volume were 11.20 mm (IQR 9.10–13.70), 9.80 mm (IQR 7.80–11.70), 4.53 mm (IQR 3.33–6.43) and 253.8 mm³ (IQR 127.5–477.0).

Local tumor control and eye retention rates were 97.7% and 86.4% after 5 years, respectively. Eight patients developed metastatic disease and 3 of them died due to metastatic disease during the follow-up period. Median visual acuity decreased from 0.67 initially to 0.05 at the last individual follow-up ($p < 0.001$).

The most common toxicities (any grade) were radiation retinopathy ($n = 39$), optic neuropathy ($n = 32$), radiogenic cataract ($n = 21$), neovascular glaucoma ($n = 15$) and dry eye syndrome ($n = 10$). The 5 year probabilities to remain free of these side effects (any grade) were 26.0%, 45.4%, 55.4%, 72.6% and 80.5%, respectively. The most important prognostic factors for toxicities were the largest tumor base diameter, tumor height and tumor distance to the optic disk.

Conclusion: Hypofractionated stereotactic photon radiotherapy with a total dose of 50 Gy delivered in 5 fractions is a highly effective treatment option in patients with centrally located choroidal melanoma and has a moderate toxicity profile.

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Modern treatment options for choroidal melanoma depend on size and location of the tumor as well as available resources. They include radiotherapy as well as various surgical techniques, with primary enucleation being required mainly in cases presenting with large tumors. Regarding brachytherapy (BT), episcleral plaques with ruthenium-106 and iodine-125 are the most widespread alternatives [3,25,27]. Alternative external beam therapy techniques include charged particle therapy using protons (or helium ions), stereotactic radiosurgery (SRS) at a Gamma Knife as well as linear accelerator (LINAC) based stereotactic radiotherapy

[4,6–9,15,17,19,20,26]. Hypofractionated stereotactic photon therapy with a LINAC has been demonstrated to be safe and effective in the treatment of uveal melanoma over the last decade [10,11,21]. The main advantages of LINAC radiotherapy are that surgical placement of tantalum markers prior to radiotherapy is not necessary when compared to proton beam therapy and a more convenient eye fixation technique than with Gamma Knife SRS.

At our institutions, hypofractionated stereotactic photon radiotherapy has been employed successfully in the treatment of choroidal melanoma. Long-term treatment results regarding local tumor control, visual acuity, metastatic disease, survival and radiogenic side effects were published earlier for patients treated with five fractions at 14, 12 and 10 Gy each, respectively [10,11].

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